
SMART INDIA HACKATHON 2026

PROBLEM STATEMENT 26173 · ISRO · DEPARTMENT OF SPACE

iTantra

*Indian Multilingual TTS & STT Aided Neural Transceiver
Radio Access for Low Bitrate Links*

Speech goes in one end. Speech comes out the other. In between it becomes a few dozen bytes — small enough to cross a radio link that could never carry a voice.

2 133×

COMPRESSION

<1 s

END TO END

10

LANGUAGES

0

INTERNET

COMPLETE TECHNICAL DESIGN DOCUMENT · REVISION 1.0

What is in this document

A complete engineering account of iTantra: the problem, the physics that makes it hard, the architecture that solves it, the numbers we will be judged on, everything that can go wrong, and the plan to build it.

PART I — THE CASE

01	The Night the Radio Stayed Silent	<i>the story</i>
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03	Compressing Meaning, Not Waveform	<i>the core idea</i>

PART II — THE SYSTEM

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B	Reference Implementation	<i>core code paths</i>

HOW TO READ THIS

Sections 01–03 are the argument — read these before building anything, and build your presentation from them. Sections 04–09 are the engineering specification. Sections 10–12 are what the jury scores. Sections 13–15 are the plan. The appendices are reference material for whoever writes the code.

The night the radio stayed silent

Every engineering decision in this document traces back to one physical fact. It is easier to understand that fact as a story than as a number.

It is 2 a.m. in the Mahanadi delta. The water has been rising for nine hours. A relief team of four is on an embankment eleven kilometres from the nearest road, and the mobile towers went down before midnight. What they carry is a handheld radio set — the kind that works when everything else has failed, because it needs no infrastructure at all, just a battery and an antenna.

The team leader needs to say eight words to base: *two people injured, we need a boat, sector seventeen*. He presses the transmit key. The link is a narrowband data channel — a few hundred bits per second, all that the terrain, the power budget and the spectrum allocation will allow.

Nothing intelligible comes out the other end. Not because the radio is broken. Because **human speech, as audio, does not fit**.

Why the audio does not fit

Sound is a demanding thing to transmit. To reproduce a human voice with enough fidelity that words remain distinguishable, a microphone must sample air pressure sixteen thousand times a second, and each of those samples needs sixteen bits. That is a quarter of a million bits for every second of speech.

Modern codecs work miracles on that figure. Opus, the best open-source speech codec available, can carry an intelligible voice at six thousand bits per second — a forty-fold reduction, at the cost of a thin, robotic timbre. It is a genuinely remarkable piece of engineering.

It is also still ten to twenty times too much for the link in the story.

256 000

BITS/SEC — RAW SPEECH

6 000

BITS/SEC — OPUS FLOOR

300

BITS/SEC — THE LINK

This is not a software problem that a cleverer codec will eventually solve. It is a bound. Audio codecs compress the *waveform* — the shape of the pressure wave — and a waveform detailed enough to be understood has an irreducible size. You can approach that bound. You cannot cross it by a factor of twenty.

The move that changes the problem

So stop transmitting the waveform.

The sentence *two people injured, we need a boat, sector seventeen* carries perhaps forty bytes of actual information. Everything else in those three seconds of audio — the speaker's timbre, the wind behind him, the exact pitch contour, the room acoustics — is **not the message**. It is the carrier the message arrived in.

If the phone in his hand converts that speech to text *before* transmission, and the phone at base converts the text back to speech *after* reception, then what crosses the radio link is forty bytes instead of ninety-six thousand. The two humans still only ever speak and listen. Neither of them reads or types anything. Between them, the message travels as language rather than as sound.

*We are not transmitting audio.
We are transmitting meaning.*

THE ENTIRE THESIS, IN EIGHT WORDS

Why this must be speech at both ends

A reasonable objection: if the message is text anyway, why not simply let people type it? The problem statement answers this directly, and the answer is the reason the project exists rather than merely being clever.

- **Literacy.** A system that requires typing excludes the people most likely to need it. In a disaster the person with critical information is a fisherman, a farmer, a village elder — not necessarily someone who can compose a message in a script under stress.
- **Hands and eyes.** A soldier at altitude wearing gloves, a rescuer on a rope, a driver — none of them can type. All of them can press one large button and talk.
- **Speed under stress.** Speaking a sentence takes three seconds. Typing it in Devanagari on a phone keyboard, in the dark, with cold hands, takes forty — if it happens at all.
- **Language.** Ten Indian languages, ten scripts. Voice is the only interface that is equally native to all of them.

So the interface is voice, because voice is inclusive. The transmission is text, because text fits. iTantra is the machine that sits between the two.

WHAT WE ARE BUILDING

An Android application that turns two ordinary phones — or a phone and a radio module — into a walkie-talkie in which speech is converted to text on the sending device, transmitted as a few dozen bytes, and converted back to speech on the receiving device. Ten Indian languages. Entirely offline. No cloud, no SIM, no proprietary SDK.

The problem, decoded

The problem statement is written in the language of a requirements document. Translated into engineering terms, it specifies eleven concrete deliverables and four hard constraints.

2.1 Requirements, restated

#	WHAT THE STATEMENT SAYS	WHAT THAT MEANS WE MUST BUILD
R1	Lightweight, highly accurate STT and TTS for ten Indian languages	Quantised on-device ASR and TTS models per language, with measured word error rate and a reported model footprint
R2	Runs locally on a low-power device	Inference on a low-end Android phone — no server, budgeted RAM, CPU and thermal envelope
R3	STT activated after detecting pauses and stoppages	Voice activity detection plus endpointing: a state machine that decides when a sentence has ended
R4	Should form the sentences detected	Segment-level finalisation, punctuation and text normalisation — not a stream of loose words
R5	Instantly and efficiently stream the data	A compact binary wire protocol with minimal framing overhead
R6	Through Wi-Fi/Bluetooth connected embedded device or another phone	A transport abstraction with at least three implementations, one of which is serial-shaped so it can drive a radio module
R7	TTS converts received text into intelligible speech, played as a voice note	Streaming synthesis with playback beginning before synthesis completes
R8	Alert messages announced at highest volume, non-interruptible	Alarm-stream audio routing, forced volume, exclusive audio focus, wake lock, lock-screen delivery
R9	Two phones, one TTS one STT, verify the complete loop	Symmetric application — every device runs both halves and switches role by mode
R10	Work like a walkie-talkie using push-to-talk	Half-duplex mode with a physical or on-screen PTT control and floor state
R11	If turned off it should work like a phone	Full-duplex mode: continuous VAD-gated streaming both directions, with barge-in

2.2 Constraints — non-negotiable

CONSTRAINT	CONSEQUENCE FOR THE DESIGN
Open source only	No proprietary voice SDK. Rules out Google Speech Services, Azure, Picovoice and similar. Also rules out Google Nearby Connections for transport — we use raw sockets instead. Every dependency is licence-audited and listed in Section 12.
Fully offline	No network call at runtime, ever. Language packs may be fetched once during setup; the operating system never touches a network. The demonstration is conducted in aeroplane mode.
Approved frameworks	TensorFlow Lite, PyTorch Mobile, ONNX Runtime and equivalents. Our runtime is ONNX Runtime via sherpa-onnx — permissively licensed, ARM-optimised, and explicitly within scope.
Low and mid-range phones	Target device is a four-gigabyte, entry-tier Snapdragon or Helio handset. All numbers in this document are to be measured on that class of hardware, not on a flagship.

2.3 The rubric is the specification

Eighty per cent of the marks are allocated to three measurable properties. This is unusual and it is a gift: the jury has told us exactly what to instrument.

WEIGHT	CRITERION	WHAT IT FORCES US TO DO
40%	Accuracy	Low word error rate for STT; high legibility and natural flow for TTS. Demands a real evaluation harness over a held-out benchmark set, per language, reported honestly — including under noise.
20%	Latency	Speech-to-text delay, text-to-audio delay, real-time factor, and the end-to-end delta between a sentence spoken on one phone and heard on the other. Demands per-stage instrumentation built into the app.
20%	Efficiency	Model size, application size, RAM and flash footprint, and CPU usage while idle-listening. Demands quantisation, on-demand language packs and a tiered wake-up path.
20%	Implied	Completeness, robustness and presentation of the delivered system.

DESIGN PRINCIPLE DRAWN FROM THE RUBRIC

Instrument from day one. The application will display a live latency and resource heads-up display, and will be able to print its own scorecard. Most competing teams will demonstrate that their system works; comparatively few will be able to show *measurements*. The rubric is asking for measurements.

Compressing meaning, not waveform

The compression ratio is the headline number of this project. It deserves to be derived carefully rather than asserted.

3.1 The arithmetic

Take a representative operational sentence in Hindi:

हमें तुरंत मदद चाहिए, दो लोग घायल हैं

“We need help immediately, two people are injured.” Duration when spoken: about 3.0 seconds.

REPRESENTATION	SIZE	BITSNOTE
Raw PCM, 16 kHz, 16-bit mono	96 000 B	768 000What the microphone produces
Opus at 16 kbps	6 000 B	48 000Good quality voice call
Opus at 6 kbps	2 250 B	18 000Thin, robotic, the practical floor
UTF-8 text	99 B	792What the recogniser produces
+ framing and addressing	111 B	88812-byte header, Section 08
+ script packing (Level 2)	45 B	360One byte per Devanagari codepoint
Template code (Level 3)	13 B	104Header plus a one-byte message ID



3.2 Three levels of compression

The system does not use one compression scheme; it degrades gracefully through three, choosing the most aggressive one the situation allows.

Level 1 — Semantic transcoding

Speech becomes UTF-8 text. This is where the enormous ratio comes from: roughly 770 to 1 before any further work. Everything below is refinement.

Level 2 — Script packing

UTF-8 encodes Devanagari, Tamil, Kannada and the other Indic scripts at three bytes per character — expensive, because UTF-8 is designed to represent all scripts at once. But the frame header already declares which language this message is in. Given that, the receiver knows exactly which codepoint block applies, and roughly seventy code points cover ordinary running text in any single Indic script. We therefore remap each script into a private single-byte alphabet, and unpack it at the far end. **Three bytes become one.** The transformation is lossless, table-driven, costs microseconds, and is unique to the fact that we control both endpoints.

Level 3 — Template codes

In an emergency, vocabulary is small and predictable. A shared table maps the most common operational sentences to single-byte identifiers.

```
0x01 We need medical assistance
0x02 Fire — evacuate immediately
0x03 Position secure, no casualties
0x04 Send a boat
0x05 Request immediate extraction
... up to 255 entries per deployment profile
```

After recognition, the resulting text is fuzzy-matched against this table. On a confident match the system transmits the identifier rather than the sentence — a single byte. The receiver's table renders the full sentence and speaks it. On a weak match it falls back to Level 2 automatically. The user is never aware of the switch.

A CONSEQUENCE WORTH NOTICING

Because each device holds the template table in *all ten languages*, a message sent as a template code is spoken in whatever language the receiver has selected. A Hindi speaker's alert reaches a Tamil speaker in Tamil — with no translation model, no additional download, and no additional latency. Cross-language operation falls out of the compression scheme for free.

3.3 What we lose, and how we compensate

Honesty about the trade-off is a strength in front of a technical jury, and every loss below has a designed answer.

WHAT IS DISCARDED	WHY IT MATTERS	COMPENSATION
Speaker's voice identity	Every message sounds like the same synthetic voice; the listener cannot tell who spoke	A one-byte sender identifier in the header, a display name in the UI, and a distinct synthesis voice per node using multi-speaker embeddings
Emotion, urgency, prosody	A shout and a whisper produce identical text	An explicit priority flag in the header; alert traffic is routed to the alarm stream at maximum volume, and the synthesis voice uses a faster, harder delivery profile
Background audio	The listener loses situational cues — gunfire, water, machinery	Accepted. Out of scope, and arguably desirable given the bandwidth budget.
Anything the recogniser gets wrong	An error is unrecoverable; the listener has no audio to fall back on	Confidence scoring transmitted with each frame, on-screen display of the recognised text before and after transmission, domain vocabulary biasing, and a low-confidence fallback to 6 kbps Opus over the Wi-Fi transport

3.4 Why this matters to ISRO specifically

The bandwidth table below is the justification for the entire project, and belongs in the presentation immediately after the compression figure.

LINK TYPE	USABLE RATE	CARRIES A VOICE	CARRIES AN ITANTRA SENTENCE?
		CALL?	
LoRa, SF10, 125 kHz	~ 1 kbps	No	Yes — 0.4 s
LoRa, long range profile	~ 0.3 kbps	No	Yes — 1.2 s
HF radio data link	~ 2 kbps	No	Yes — 0.2 s
Narrowband satellite burst	~ 1 kbps	No	Yes — 0.4 s
Bluetooth RFCOMM	~ 200 kbps	Marginally	Yes — 2 ms
Wi-Fi Direct	> 10 Mbps	Yes	Yes — instant

Read the middle column, then the right. On the four links that matter operationally — the ones that work when infrastructure does not — audio is impossible and iTantra is comfortable. That gap is the product.

System architecture

Every device runs the complete system. Nothing is a dedicated transmitter or a dedicated receiver; role is a runtime mode, not a build.

4.1 End-to-end signal path

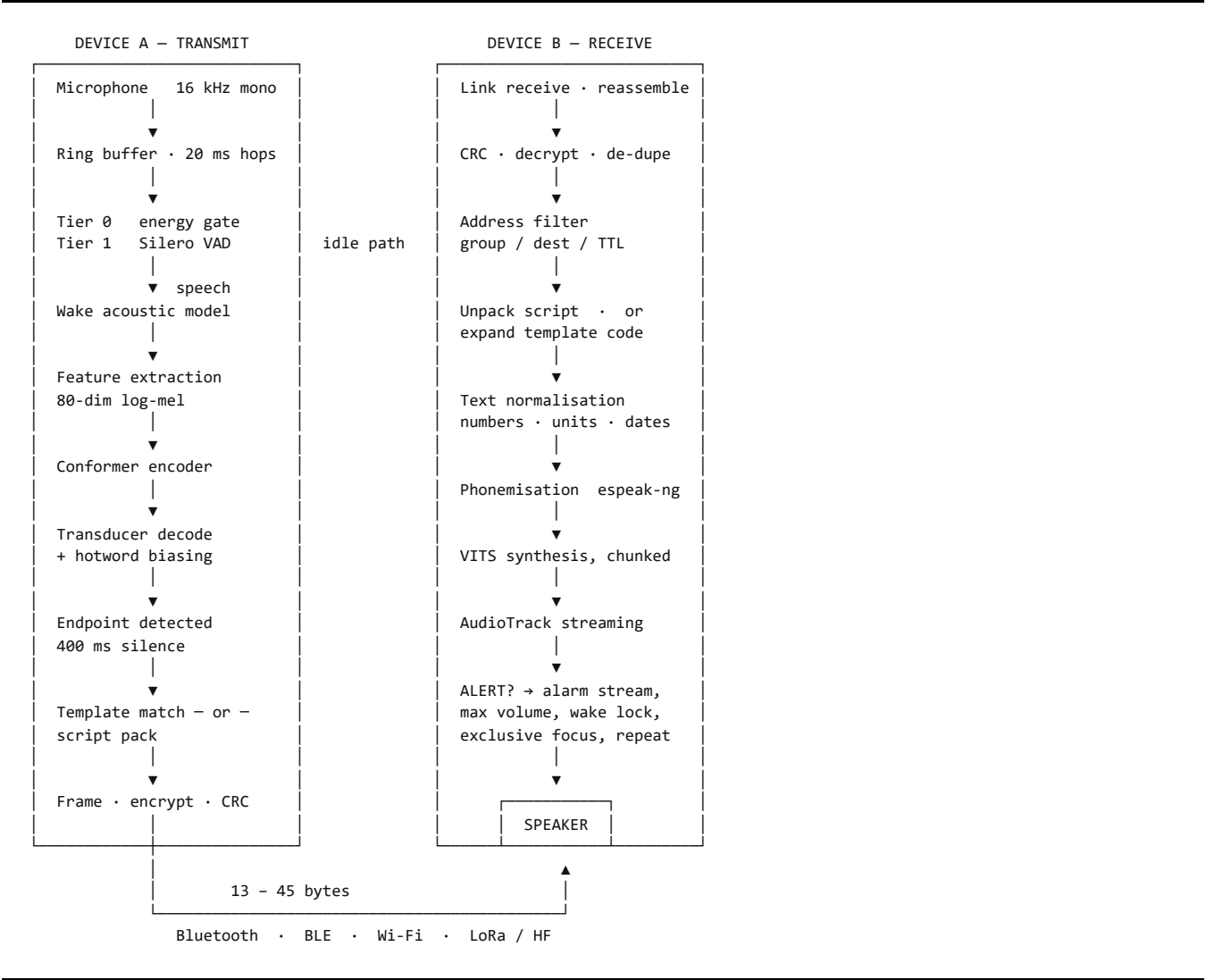


FIGURE 4.1 — COMPLETE SIGNAL PATH. BOTH HALVES ARE PRESENT ON EVERY DEVICE.

4.2 Module map

MODULE	RESPONSIBILITY	PRINCIPAL CONTENTS
app	User interface and orchestration	Compose screens, PTT control, channel and roster UI, settings, latency HUD, provisioning by QR
core-audio	Capture and playback	AudioRecord and AudioTrack wrappers, ring buffers, gain control, noise suppression, alert audio policy, focus handling
core-asr	Speech to text	sherpa-onnx recogniser binding, three-tier VAD, endpointing state machine, hotword and gazetteer biasing, confidence scoring
core-tts	Text to speech	Per-language normalisation rules, phonemiser, chunked VITS synthesis, streaming playout scheduler
core-link	Transport	Link interface, RFCOMM, BLE GATT, Wi-Fi socket and Serial implementations; discovery, reconnection, queueing
core-proto	Wire format	Frame encode and decode, CRC-16, AES-GCM, script packing tables, template tables, sequence and replay windows, relay logic
core-models	Asset management	Language pack manifest, checksum verification, resumable download, storage accounting, model lifecycle
bench	Evaluation	WER harness, RTF measurement, latency logger, noise mixing, CSV and scorecard export

4.3 Threading and lifecycle

Audio work must never touch the user-interface thread, and the recogniser must never block capture. Four threads, with bounded queues between them:

THREAD	PRIORITY	DUTY
Capture	URGENT_AUDIO	Reads 20 ms blocks from AudioRecord, runs the energy gate and VAD, pushes speech blocks into the ASR queue. Must never allocate.
Inference	AUDIO	Drains the ASR queue, runs encoder and decoder, emits partial and final hypotheses.
Link	DEFAULT	Blocking socket reads and writes, framing, retransmission timers, heartbeat.
Synthesis	AUDIO	Chunked TTS generation feeding an AudioTrack in streaming mode.

The whole engine lives in a foreground service with a persistent notification, so that Android does not reclaim it while the screen is off — which is the normal operating state for a radio.

COLD START

Loading the acoustic and synthesis models costs one to two seconds. That delay must never be paid when the user presses transmit. Models are loaded when the service starts and held resident; the idle path keeps them warm but does no inference.

The listening engine

Speech recognition carries the largest share of the accuracy mark and the largest share of the compute budget. It is also where the idle-power score is won or lost.

5.1 Three-tier voice activity detection

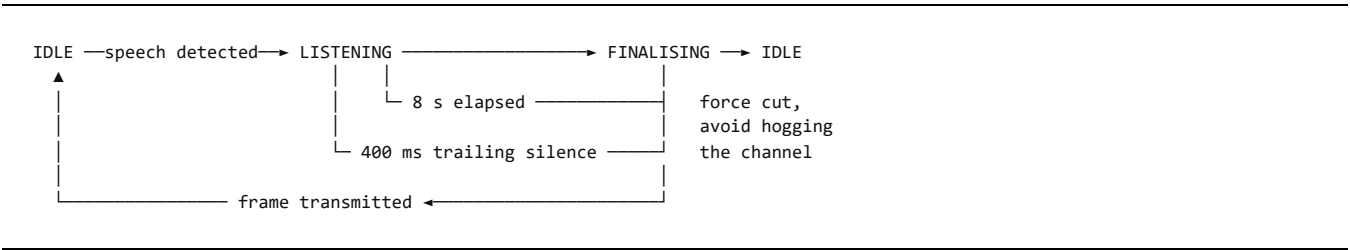
Running the acoustic model continuously would exhaust the battery in about two hours and would fail the efficiency criterion outright. Detection is therefore tiered, so that the expensive stage runs only when the cheap stages agree that a human is speaking.

TIER	MECHANISM	COST PER 20 MSPURPOSE
0	Energy gate — sum of squares against an adaptive noise floor	~ 3 μ sRejects true silence, which is the overwhelming majority of elapsed time. Effectively free.
1	Silero VAD, 1.8 MB ONNX	~ 300 μ sDistinguishes human speech from door slams, engines, wind and music. Prevents false wake-ups.
2	Conformer acoustic model	~ 4 msActual recognition. Runs only while tier 1 reports speech.

With a duty cycle typical of radio use — speech perhaps five per cent of the time — measured idle CPU should sit below two per cent of one core. That figure is to be measured with Perfetto on the target handset and reported.

5.2 Endpointing — deciding when a sentence has ended

The problem statement asks for activation “after detecting pauses and stoppages” and for the system to “form the sentences detected”. This is the endpointing state machine.



PARAMETER	VALUATIONALE
Trailing silence, phone mode	400 msShorter clips natural inter-clause pauses; longer is perceptible as lag
Trailing silence, PTT mode	150 msReleasing the key is itself an explicit end-of-utterance signal
Leading pad	250 msRetained from the ring buffer so the first phoneme is never clipped
Maximum utterance	8 sBounds latency and prevents one speaker monopolising a half-duplex channel
Minimum utterance	300 msSuppresses coughs, clicks and key noise

THE SINGLE LARGEST LATENCY TERM

The 400 ms silence window is the biggest component of end-to-end delay, and it is a design choice rather than a computational cost. In push-to-talk mode the key release replaces it entirely, which is why PTT mode is measurably faster than phone mode — a point worth demonstrating live.

5.3 From pressure wave to text

Feature extraction

Raw samples are an inefficient representation. Audio is cut into 25 ms windows every 10 ms; each window is transformed to the frequency domain and projected onto 80 mel bands, which are spaced the way human hearing resolves pitch. Three seconds of speech becomes a 300×80 matrix — a compact spectro-temporal fingerprint. Cost is roughly one millisecond, computed in native code.

Acoustic model

The feature matrix is consumed by a Conformer encoder: convolutional blocks that detect local acoustic events such as plosive bursts and formant transitions, interleaved with self-attention blocks that supply sentence-level context. Output is a sequence of 512-dimensional frame embeddings. This stage is approximately eighty-five per cent of the inference cost.

Decoding

A transducer or CTC head projects each frame onto a vocabulary of roughly one thousand sub-word tokens defined in the language's token table. Beam search selects the most probable token sequence; repeated and blank emissions collapse, and the surviving tokens are joined into text.

Contextual biasing — the highest-yield accuracy work

Decoding scores are boosted for a supplied phrase list. In a distress context the critical vocabulary is small and known in advance: *मदद, घायल, आग, निकासी*, unit callsigns, sector numbers, local place names. Loading a domain lexicon and a deployment gazetteer measurably reduces error on precisely the words whose misrecognition would be most costly, and requires no retraining. Demonstrating the same audio decoded with biasing off and on is one of the strongest live comparisons available to us.

5.4 Model selection

CANDIDATE	LICENCE	SIZE, INT8ASSESSMENT
AI4Bharat IndicConformer	Permissive	~ 35 MB Production choice. Trained specifically on Indian languages; covers all ten targets. Requires NeMo-to-ONNX export.
Vosk small models	Apache-2.0	~ 40 MBPrototype choice for week one. Streaming, Android-ready today, lower accuracy. De-risks the schedule.
Whisper tiny / base	MIT	~ 40–75 MBBaseline for comparison only. Not streaming; weak on Odia and Kannada; slow on entry-tier silicon.
IndicWhisper	Permissive	~ 240 MBStrong accuracy, too heavy for the target device. Useful as an accuracy ceiling reference.
Meta MMS	CC-BY-NC	~ 300 MBBroadest coverage; non-commercial licence makes it unsuitable for a deployable claim. Gap-filler only.

5.5 Quantisation

Models are trained in 32-bit floating point, which is unnecessary for inference. Dynamic quantisation rewrites the weights as 8-bit integers, exploiting the fact that ARM cores execute integer arithmetic considerably faster than float.

PRECISION	SIZE	RELATIVE SPEED	WER IMPACTVERDICT
float32, as trained	~ 120 MB	1.0×	baselineUnshippable
float16	~ 60 MB	1.3×	≈ 0Intermediate
int8 dynamic	~ 32 MB	2.5×	+ ~1% rel. Shipped

Four times smaller, two and a half times faster, for approximately one per cent relative degradation. This single transformation is what makes on-device speech viable on an entry-tier handset, and the before-and-after table is direct evidence for the efficiency criterion.

```
python -m onnxruntime.quantization.preprocess --input model.onnx --output prep.onnx

python - <<'PY'
from onnxruntime.quantization import quantize_dynamic, QuantType
quantize_dynamic("prep.onnx", "model.int8.onnx", weight_type=QuantType.QInt8)
PY
```

The speaking engine

Synthesis is scored on intelligibility and natural flow. Two of the four stages below are ordinary software engineering rather than machine learning, and they account for most of the perceived quality.

6.1 Text normalisation

No neural synthesiser handles raw digits, units or abbreviations correctly, and these are exactly what operational messages contain. A per-language rule file runs before synthesis.

INPUT	NAÏVE OUTPUT	CORRECT OUTPUT
112	one hundred and twelve	एक · एक · दो — digit-wise, because it is a number to dial
5 km	five k m	पाँच किलोमीटर
Sector 17	sector seventeen	सेक्टर सत्रह
14:30	fourteen colon thirty	दोपहर दो बजकर तीस मिनट

Distinguishing a quantity from an identifier is a rule, not a model: bare digit runs following a dialling or callsign context are read digit-wise, quantities followed by a unit are read as numbers. Cheap to implement, and it is the difference between a synthesiser that sounds professional and one that sounds broken.

6.2 Phonemisation

Letters are not sounds. `espeak-ng` converts orthography to phoneme sequences — मदद to /m ə d ə d/ — which are then mapped to integer identifiers. Indic scripts are largely phonetic, which makes this stage far more reliable for our ten languages than it would be for English.

LICENCE NOTE

`espeak-ng` is GPL-3.0. This is acceptable for an open-source submission and is recorded in the licence inventory, but it constrains any future closed-source derivative and must be disclosed rather than discovered.

6.3 VITS synthesis

Earlier architectures required two models — text to spectrogram, then spectrogram to waveform — and were too slow for phones. VITS performs both in a single pass:

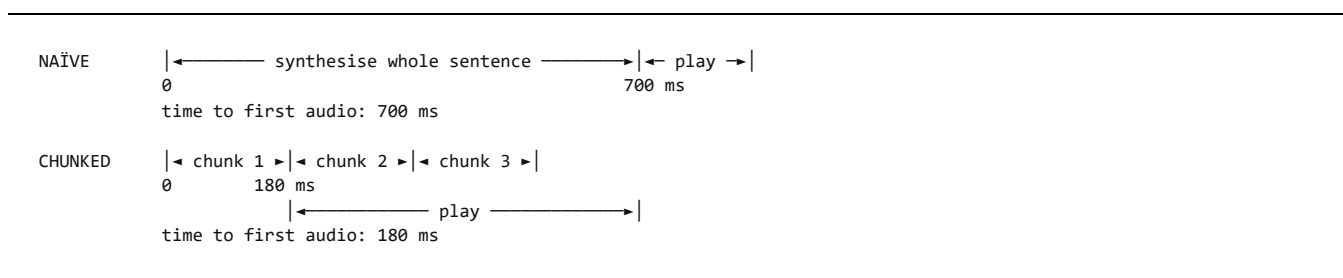
— A **text encoder** transforms phoneme identifiers into latent representations.

- A **duration predictor** assigns a length to each phoneme. This is the stage that produces natural rhythm, and it is why VITS output does not sound metronomic.
- A **flow-based decoder and vocoder** generates the waveform directly at 22 050 samples per second.

Measured real-time factor on entry-tier ARM is approximately 0.15 — one second of speech synthesised in about 150 milliseconds.

6.4 Chunked synthesis and streaming playback

Synthesising a complete sentence and then playing it wastes the entire synthesis duration as latency. Instead the sentence is split at clause boundaries; the first chunk is synthesised and playback begins immediately while subsequent chunks are generated against the playing clock.



Total synthesis work is unchanged; perceived latency falls by roughly seventy-five per cent. Since the rubric measures “time delay between the text received and audio processed and played”, this optimisation is directly worth marks.

6.5 Voice selection

SOURCE	LICENCE	PER VOICEASSESSMENT
Piper voices	MIT	20–60 MBPrimary. Already ONNX, fast, and covers nine of the ten targets.
AI4Bharat Indic-TTS	Permissive	~ 90 MBBest Indic naturalness. Reserve for languages where Piper quality is inadequate; quantise aggressively.
Coqui VITS, self-trained	MPL-2.0	~ 50 MBTraining path for gaps. Odia is the expected gap; IIT Madras IndicTTS data supports this.
Meta MMS-TTS	CC-BY-NC	~ 40 MBStopgap only, owing to the non-commercial licence.
Matcha-TTS	MIT	~ 35 MBFlow-matching alternative; benchmark against Piper for RTF before committing.

The transport layer

Four physical media, one interface. The application never learns which one is carrying its bytes.

7.1 The abstraction

```
interface Link {
    suspend fun send(frame: ByteArray)
    val incoming: Flow<ByteArray>
    val state: StateFlow<LinkState>      // IDLE · DISCOVERING · CONNECTED · DEGRADED · ERROR
    val mtu: Int
    val metrics: LinkMetrics             // rssi, round-trip, loss, queue depth
}
```

Four implementations satisfy this contract. Changing transport is a settings toggle, and adding one costs a single file. This abstraction is also what allows the entire transport layer to be developed and tested in week one, before any model exists — a text field replaces the recogniser, and the whole path is exercised.

7.2 Bluetooth Classic — RFCOMM

RFCOMM is a virtual serial cable. It is the primary transport for two reasons: it is a reliable ordered byte stream requiring no additional protocol work, and it is the *same* profile that an ESP32 or a radio modem exposes. The code that talks to another phone is the code that talks to a radio.

```
val ITANTRA_UUID = UUID.fromString("8ce255c0-200a-11e0-ac64-0800200c9a66")

// host
val server = adapter.listenUsingRfcommWithServiceRecord("iTantra", ITANTRA_UUID)
val socket = server.accept(); server.close()

// peer
adapter.cancelDiscovery()                // discovery cripples throughput
val socket = device.createRfcommSocketToServiceRecord(ITANTRA_UUID).apply { connect() }
```

7.3 Bluetooth Low Energy — GATT

BLE is not a stream but a set of addressable characteristics. One characteristic is written by the peer, one is notified by the host. Default MTU is 23 bytes; negotiating 247 yields roughly 244 usable, so a typical sentence fits in a single packet.

Power consumption is roughly a tenth of Bluetooth Classic, which makes BLE the correct standby transport and supports the multi-hour battery claim. It is also natively broadcast, which makes group operation straightforward.

7.4 Wi-Fi — Direct or hosted network

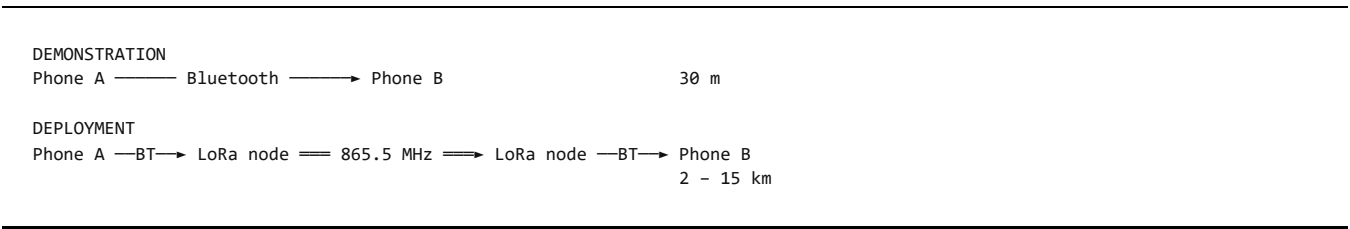
Two mechanisms produce the same result: an isolated local network between the devices, with no router, no SIM and no internet. Wi-Fi Direct negotiates a group automatically through `WifiP2pManager`; a hosted network — the ordinary hotspot toggle with mobile data off — achieves the same thing with far greater reliability across manufacturers. Once either exists, transport is plain TCP.

ENGINEERING RECOMMENDATION

Prefer the hosted-network path. `WifiP2pManager` discovery is inconsistent across Samsung, Xiaomi and Realme builds and consumed a substantial fraction of many teams' schedules; the hosted network requires no P2P API at all. Peer discovery is then a single UDP broadcast on the subnet.

7.5 Serial — the deployment path

The fourth implementation sends identical frames over Bluetooth SPP to a radio module. The phone is unchanged; only what sits at the far end of the Bluetooth connection differs. This is the configuration the problem statement describes when it refers to a “Wi-Fi/Bluetooth connected embedded device”.



7.6 Comparison

PROPERTY	BT CLASSIC	BLE	WI-FI	LORA SERIAL
Range	10–30 m	10–50 m	50–150 m	2–15 km
Usable rate	~200 kbps	5–20 kbps	>10 Mbps	0.3–5 kbps
Power	Medium	Very low	High	Low
Model	Byte stream	Packets	Byte stream	Packets
Native broadcast	No	Yes	Multicast	Yes
Audio fallback viable	Marginal	No	Yes	No
Role	Default	Standby	Range & fallback	Deployment

7.7 Framing over a stream

RFCOMM and TCP preserve byte order but not message boundaries. Writing 45 bytes and then 38 may be read as 20 bytes and then 63. Every frame therefore carries an explicit length, and every read loops until the declared number of

bytes has arrived.

```
fun InputStream.readFully(buf: ByteArray) {  
    var off = 0  
    while (off < buf.size) {  
        val n = read(buf, off, buf.size - off)  
        if (n < 0) throw EOFException()  
        off += n  
    }  
}
```

Omitting this loop produces a system that works on a desk and fails under load. It is the single most common defect in student implementations of this class of project.

The wire protocol

Twelve bytes of header buy addressing, channels, priority, relay, integrity and authentication. The payload is what remains.

8.1 Frame layout

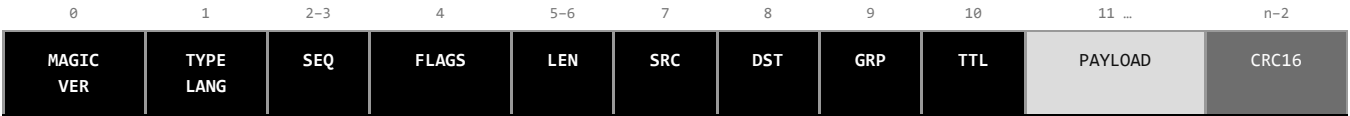


FIGURE 8.1 — FRAME LAYOUT. HEADER 11 BYTES, TRAILER 2 BYTES.

FIELD	BYTESDEFINITION
MAGIC / VER	1High nibble 0xA for resynchronisation after corruption; low nibble protocol version
TYPE / LANG	1High nibble: TEXT, ALERT, ACK, PTT_CTL, HEARTBEAT, TEMPLATE, AUDIO_FALLBACK. Low nibble: language index 0–9
SEQ	2Per-sender sequence number; drives acknowledgement, duplicate suppression and the replay window
FLAGS	1FINAL, PARTIAL, ENCRYPTED, FRAGMENT, plus two bits of recogniser confidence
LEN	2Payload length; the basis of stream framing
SRC	1Sender node identifier
DST	1Destination node, or 0xFF for broadcast
GRP	1Channel identifier — the equivalent of a walkie-talkie channel
TTL	1Remaining relay hops; decremented on forward, dropped at zero
PAYLOAD	nPacked text, template identifier, coordinates, or control data — encrypted when the flag is set
CRC16	2CCITT over header and payload

8.2 Addressing — how the message reaches the right listener

Radio does not route; it broadcasts. Every device within range receives every transmission, and selection happens at the receiver. This is how aviation, military and amateur radio have always worked, and iTantra reproduces it in software.

```

fun accept(f: Frame): Boolean = when {
    f.grp != myGroup      -> false    // different channel – ignore silently
    f.dst == BROADCAST    -> true     // addressed to all units
    f.dst == myNode       -> true     // addressed to me
    else                  -> false    // someone else's traffic
}

```

Three bytes therefore deliver broadcast operation, private one-to-one messages, and independent channels sharing the same physical medium.

Relay

Where a device cannot reach its destination directly, intermediate devices forward the frame and decrement TTL. A seen-set of recent source and sequence pairs suppresses duplicates; without it, three mutually visible devices generate an unbounded broadcast storm.

```

if (f.dst != myNode && f.ttl > 0 && seen.add(f.src to f.seq))
    link.send(f.copy(ttl = f.ttl - 1))

```

8.3 Security

Address filtering is a convention, not a control — any device can set any destination byte. In a system whose purpose is to raise alarms, the ability to inject a fraudulent maximum-volume evacuation order is a weapon. Three measures are mandatory.

MEASURE	PREVENTS	IMPLEMENTATION
AES-256-GCM per group	Eavesdropping and forgery	Pre-shared key distributed at provisioning. A device outside the group receives the radio signal, fails the authentication tag, and discards the frame without ever learning its contents. Encryption is the real address.
Sliding replay window	Capture and retransmission	A 64-entry window per sender; a sequence number already seen, or older than the window, is rejected. Without this, a recorded alert can be replayed indefinitely.
Bound sender identity	Impersonation	The source identifier is inside the authenticated region, so it cannot be altered without invalidating the tag.

8.4 Reliability

MECHANISM	APPLIES TO	BEHAVIOUR
CRC-16	All frames	Corrupt frames are discarded, never spoken. A garbled instruction is more dangerous than a missing one.
Acknowledge and retry	ALERT only	Up to three retries at 300 ms. Ordinary conversation is fire-and-forget — retransmitting stale speech is worse than losing it.
Heartbeat	Link	Every 2 s, so the interface can distinguish a quiet channel from a dead one.
Store and forward	All	Frames queue in a local database when the peer is unreachable and flush on reconnection, with pending, sent and delivered states surfaced in the UI.
Reconnection	Link	Exponential backoff with jitter; the service restores the socket without user action.

8.5 Payload budget

FRAME	HEADER	PAYLOAD	TOTAL	ON 300 BPS LORA
Template alert with coordinates	13 B	9 B	22 B	0.6 s
Template alert	13 B	1 B	14 B	0.4 s
Packed Hindi sentence	13 B	32 B	45 B	1.2 s
Plain UTF-8 sentence	13 B	99 B	112 B	3.0 s
Equivalent Opus audio	—	2 250 B	2 250 B	60 s

The application

The justification for this project is that it serves people who cannot type. An interface that assumes literacy would refute its own premise.

9.1 Two modes

ASPECT	PUSH-TO-TALK — WALKIE-TALKIE	RELEASED — TELEPHONE
Duplex	Half. One speaker holds the channel.	Full. Both directions stream continuously.
Capture	Live only while the key is held	Always live, gated by VAD
Endpoint	Key release, 150 ms confirmation	400 ms trailing silence
Speaker	Muted while transmitting	Active, with barge-in ducking
Floor control	PTT_CTL frames announce and release the floor; a busy indicator prevents collisions	Not applicable
Power	Lowest — no idle inference	Higher — continuous VAD

The transmit control is bound both to a large on-screen target and to the volume-down hardware key, because operators wear gloves and rarely look at the screen. Releasing the key is an explicit end-of-utterance signal, which is why push-to-talk mode records lower latency than telephone mode.

9.2 Alert delivery

The requirement is that alerts are “announced at highest volume, non-interruptible”. On Android this is a specific and verifiable sequence.

STEP	MECHANISM
Route away from media	<code>AudioAttributes.USAGE_ALARM</code> with <code>CONTENT_TYPE_SONIFICATION</code> — bypasses media volume and, when configured, Do Not Disturb
Force volume	<code>setStreamVolume(STREAM_ALARM, max, 0)</code> before playback; the prior level is restored afterwards
Hold focus	Request <code>AUDIOFOCUS_GAIN_TRANSIENT_EXCLUSIVE</code> and deliberately ignore loss callbacks for alert priority — this is the non-interruptible requirement
Wake the device	<code>PARTIAL_WAKE_LOCK</code> plus a full-screen-intent notification so delivery succeeds on a locked screen
Reinforce	Vibration pattern, high-contrast full-screen visual, message repeated twice
Guarantee arrival	ALERT frames pre-empt the transmit queue and are acknowledged and retried

9.3 Interface

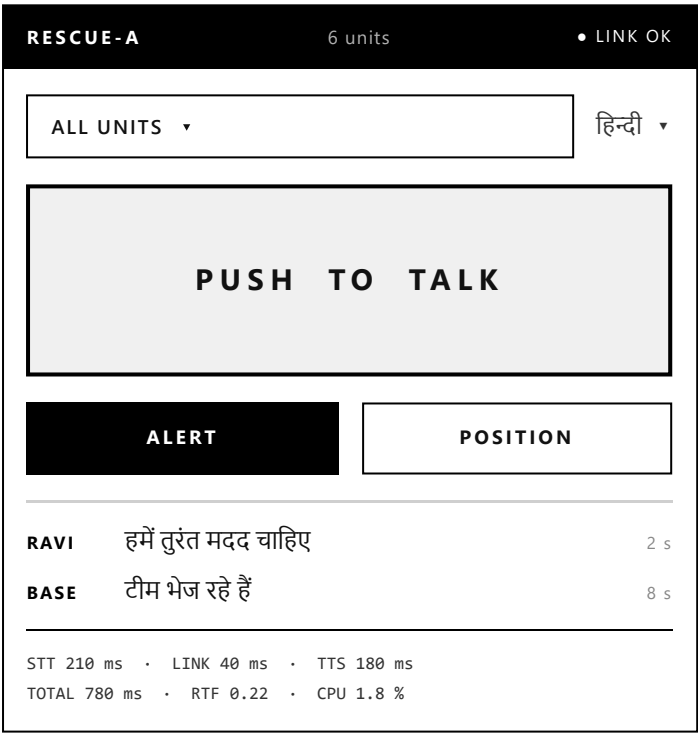


FIGURE 9.1 — OPERATING SCREEN. THE INSTRUMENTATION STRIP IS PERMANENT, NOT A DEBUG VIEW.

Inclusive design rules

- The transmit target occupies at least a third of the screen and is reachable one-handed with gloves.
- Every state change is confirmed by haptics and by a spoken cue, never by a text dialog alone.
- Icons and colour carry primary meaning; text is a secondary channel.
- Full operation with the screen off, via the hardware key.
- High-contrast monochrome palette legible in direct sunlight.
- Recognised text is displayed alongside the spoken output, so a literate operator can verify what the machine heard — without requiring literacy to use the system.

9.4 Provisioning

Users are not asked to enter identifiers. One device creates the group and generates a random channel identifier, an AES key and node assignments; it displays a QR code. Joining devices scan it, receive the key, and claim the next free node identifier. Display names are local. Pairing is the most common point of failure in live demonstrations, and reducing it to a scan removes that risk.

9.5 Language packs

Shipping ten languages in the installer would produce a package of roughly seven hundred megabytes and forfeit the efficiency criterion. Instead assets are modular.

COMPONENT	SIZEDELIVERY
Base application, runtime, VAD	~ 25 MBInstaller
Hindi and English packs	~ 120 MBInstaller — usable immediately
Each additional language	~ 60 MBOn demand, once, at setup; verified by checksum
Template and gazetteer tables	< 1 MBInstaller, all languages

A pack is a manifest entry plus four files — acoustic model, token table, synthesis voice, normalisation rules. Adding a language requires no code change. Downloads occur once during setup; runtime operation is unconditionally offline.

Metrics and evaluation

Eighty per cent of the assessment is measurable. This section defines what we measure, how, and what result we consider a pass.

10.1 Efficiency — 20 per cent

QUANTITY	TARGETMETHOD
Installer size	< 30 MBarm64-v8a split, App Bundle, no bundled weights beyond the base pack
Per-language footprint	< 70 MBint8 quantised acoustic model and voice
Resident memory, one language	< 350 MBAndroid Studio memory profiler under sustained load
Idle CPU, listening	< 2 %Perfetto trace over ten minutes of silence with VAD active
Active CPU, recognising	< 35 %Perfetto, single core, during continuous speech
Standby endurance	> 8 hBatterystats, BLE transport, screen off, listening
Cold start to ready	< 2 sInstrumented service start; models preloaded thereafter

10.2 Accuracy — 40 per cent

Recognition

Word error rate is measured per language on held-out benchmark data, not on self-recorded samples. Evaluation sets: AI4Bharat IndicSUPERB and Kathbath for Indic coverage, IndicVoices for spontaneous rather than read speech, and FLEURS for figures comparable with published literature.

Crucially, error rate is reported **at four signal-to-noise ratios**. Clean-room figures are not representative of a flood embankment, and a jury may test in a noisy hall. Noise is mixed from a standard set — crowd, wind, engine, siren.

CONDITION	TARGET WERCOMMENT
Clean, read speech	< 12 %Reference figure
+20 dB SNR	< 15 %Quiet room, realistic
+10 dB SNR	< 22 %Vehicle, crowd
+5 dB SNR	< 32 %Adverse; template fallback engages
Critical vocabulary, biased	< 6 %The figure that actually matters operationally

The final row is the important one. Overall word error rate is a poor proxy for usefulness; what matters is whether मद्द, अ॥, sector numbers and callsigns survive. Reporting critical-term error separately, with and without contextual biasing, demonstrates a level of evaluation rigour that general WER alone does not.

Synthesis

MEASURE	TARGETMETHOD
Mean opinion score	> 3.8 / 5Blind listening panel, minimum fifteen native speakers per language
Intelligibility	> 95 %Transcription accuracy of synthesised output by native listeners
Normalisation correctness	100 %Fixed regression suite of numbers, units, times and callsigns per language

10.3 Latency — 20 per cent

STAGE	BUDGETNOTE
Capture and buffering	20–40 ms20 ms hop size
Endpoint silence window	150–400 msLargest term; a design choice, not compute. 150 ms in PTT mode.
Final decode after endpoint	100–250 msScales with utterance length
Framing, encryption, transmit	20–60 msBluetooth; sub-millisecond of that is the payload itself
Normalisation	< 10 msRule-based
First synthesis chunk	150–250 msChunked; not full-sentence synthesis
Output pipeline	30–80 msAudioTrack; Oboe if this proves material
End to end, PTT mode	500–800 msThe figure the jury will time
End to end, phone mode	750–1100 msIncludes the silence window
Real-time factor, ASR	< 0.3oSustained, after thermal soak
Real-time factor, TTS	< 0.25Sustained

SUSTAINED, NOT PEAK

Entry-tier handsets throttle after roughly ten minutes of continuous inference, and real-time factor can double. All latency figures are to be taken after a thirty-minute soak, not from a cold device. Reporting sustained rather than peak numbers is both more honest and more defensible under questioning.

10.4 Instrumentation

Every stage boundary emits a timestamp; the application maintains a rolling histogram and can export a complete run as CSV. The operating screen displays the current figures permanently. The `bench` module runs the full evaluation offline

and emits the scorecard below.

ARTEFACT	CONTENTS
scorecard.csv	Per language: WER at four SNRs, critical-term error, TTS MOS, model sizes, RTF
latency.csv	Per utterance: every stage boundary, transport, payload size, end-to-end delta
resource.csv	Sampled CPU, resident memory, battery drain, thermal state

Producing these three files automatically converts the presentation from a set of claims into a set of results.

Issues, risks and **mitigations**

Twenty identified failure modes, each with a designed response. Severity reflects impact on the assessed criteria, not merely on the software.

11.1 Technical risks

ID	RISK	SEVERITY	MITIGATION
T-01	Word error rate collapses in real acoustic noise, destroying the largest single mark	HIGH	RNNNoise or DTLN pre-processing at roughly 200 KB; adaptive gain; 80 Hz high-pass; platform NoiseSuppressor. Evaluate and report at four SNRs rather than clean only. Template fallback engages when confidence drops.
T-02	Digits, callsigns and place names misrecognised — precisely the tokens that carry operational meaning	HIGH	Contextual biasing with a domain lexicon and deployment gazetteer; digits read and confirmed on screen before transmission; separate critical-term error metric tracked throughout.
T-03	Thermal throttling doubles real-time factor during a long demonstration	HIGH	Thirty-minute soak testing from week four; report sustained figures; reduce thread count under thermal pressure; BLE standby to lower baseline temperature.
T-04	Total asset size defeats the efficiency criterion	MED	int8 quantisation throughout; on-demand language packs; ABI splitting; a 25 MB installer with two languages resident.
T-05	No adequate open model exists for Odia	MED	Identified as the expected gap. Fallbacks in order: MMS, then a Coqui VITS voice trained on IIT Madras IndicTTS data. Assign in week four, not week seven.
T-06	Code-mixed Hindi and English speech is mis-decoded	MED	English retained in the biasing lexicon; acknowledged openly as a limitation with a stated roadmap rather than concealed. Honest scoping is defensible; a broken demonstration is not.
T-07	NeMo to ONNX export of IndicConformer fails or degrades accuracy	MED	Vosk models integrated in week one keep the schedule intact regardless. Export is attempted in week four with three weeks of slack behind it.
T-08	Stream framing defect — partial reads corrupt frames under load	MED	Mandatory readFully loop; CRC on every frame; magic-byte resynchronisation; fuzz testing of the decoder with truncated and interleaved input.
T-09	Wi-Fi Direct discovery unreliable across manufacturers	MED	Hosted-network path adopted as primary Wi-Fi mechanism; WifiP2pManager treated as optional. Bluetooth remains the default transport regardless.
T-10	Broadcast storm when three or more devices relay	LOW	TTL decrement plus a bounded seen-set of source and sequence pairs; verified with a four-device soak.
T-11	Cold-start model load stalls the first transmission	LOW	Models loaded at foreground-service start and held resident; the transmit control is disabled until ready, with a visible ready indicator.
T-12	Microphone unavailable — an incoming call or another application holds it	LOW	Audio focus and recording-configuration callbacks; explicit degraded state surfaced in the UI; automatic recovery on release.

11.2 Security and safety risks

ID	RISK	SEVERITY	MITIGATION
S-01	Fraudulent alert injected by an unauthorised transmitter, played at maximum volume	HIGH	AES-256-GCM with a per-group pre-shared key; the sender identifier lies inside the authenticated region; frames failing authentication are discarded silently.
S-02	Recorded alert replayed later	HIGH	64-entry sliding replay window per sender; sequence numbers outside the window rejected.
S-03	Recognition error inverts meaning — “do not evacuate” becomes “now evacuate”	HIGH	Confidence transmitted with every frame; recognised text displayed to the sender before transmission; explicit confirmation required for alert-class messages; negation terms added to the biasing lexicon.
S-04	Key material exposed on a captured device	MED	Android Keystore storage; group re-keying supported; a device can be removed from a group by rotating the key at the next provisioning.
S-05	Two operators transmit simultaneously on a half-duplex channel, garbling both	MED	PTT_CTL floor announcement, channel-busy indicator, randomised backoff before retry.

11.3 Programme risks

ID	RISK	SEVERITY	MITIGATION
P-01	Effort consumed by model work, leaving the end-to-end loop unproven late in the schedule	HIGH	The loop closes in week three using the weakest acceptable models. Model quality is an upgrade path, never a prerequisite. This is the single most important scheduling decision in the plan.
P-02	Development on flagship hardware conceals performance failures	HIGH	An entry-tier target handset is acquired in week one and every benchmark is taken on it.
P-03	Live pairing fails on stage	MED	QR provisioning; devices pre-paired before the session; a rehearsed recovery path; the demonstration script fixed by week eight.
P-04	Licence incompatibility discovered late	LOW	LICENSES.md maintained from week one; GPL and non-commercial components flagged on entry, not on discovery.

Open-source inventory

Every component, its licence, and its role. The submission rules prohibit proprietary voice SDKs; this table is the evidence of compliance.

12.1 Runtime and inference

COMPONENT	LICENCE	ROLE
k2-fsa / sherpa-onnx	Apache-2.0	Principal dependency. Recognition, synthesis, VAD and endpointing in one library, with an Android archive and Kotlin bindings.
microsoft / onnxruntime	MIT	Inference engine beneath sherpa-onnx; XNNPACK acceleration on ARM.
google-ai-edge / LiteRT	Apache-2.0	Named in the problem statement; evaluated as an alternative backend.
pytorch / executorch	BSD-3	Named in the problem statement; evaluated.
ggml-org / whisper.cpp	MIT	Accuracy baseline for comparison.

12.2 Recognition models

COMPONENT	LICENCE	ROLE
AI4Bharat IndicConformer	Permissive	Production acoustic models, all ten languages.
alphacep / vosk-api	Apache-2.0	Week-one prototype; schedule insurance.
NVIDIA / NeMo	Apache-2.0	ONNX export tooling and fine-tuning recipes.
k2-fsa / icefall	Apache-2.0	Streaming Zipformer training recipes, if fine-tuning is undertaken.
Meta MMS	CC-BY-NC 	Coverage gap-filler. Non-commercial — disclosed, and excluded from any deployability claim.

12.3 Synthesis

COMPONENT	LICENCE	ROLE
rhasspy / piper & piper-voices	MIT	Primary synthesis voices; already ONNX.
espeak-ng	GPL-3.0 A	Phonemisation. Copyleft — disclosed.
idiap / coqui-ai-TTS	MPL-2.0	Training path for languages without an adequate voice.
AI4Bharat Indic-TTS	Permissive	Higher-naturalness Indic voices.
shivammehta25 / Matcha-TTS	MIT	Alternative architecture; benchmarked against Piper.

12.4 Detection, transport and utilities

COMPONENT	LICENCE	ROLE
snakers4 / silero-vad	MIT	Tier-1 voice activity detection.
WebRTC VAD	BSD-3	Reference for the tier-0 energy gate.
xiph / rnnoise	BSD-3	Noise suppression ahead of recognition.
google / oboe	Apache-2.0	Low-latency audio path, if AudioTrack proves insufficient.
facebook / zstd	BSD-3	Dictionary compression above script packing.

Transport uses only the Android platform Bluetooth and socket APIs. No third-party connectivity SDK is used, and Google Nearby Connections is deliberately excluded.

12.5 Reference designs studied

PROJECT	LICENCE	WHAT IT INFORMS
meshtastic / firmware & Android	GPL-3.0	The closest existing system: text messaging over LoRa with an Android client across Bluetooth. Informs framing, addressing, relay and duplicate suppression.
Briar	GPL-3.0	Robust Bluetooth and Wi-Fi transport on Android: discovery, reconnection, degraded states.
georg / aprsdroid	GPL-3.0	Android to amateur radio over Bluetooth serial — the deployment path in production use.
android / connectivity-samples	Apache-2.0	Platform boilerplate for RFCOMM and Wi-Fi Direct.

12.6 Evaluation data

DATASET	SOURCE	USE
IndicSUPERB / Kathbath	AI4Bharat	Primary word-error-rate benchmark across all ten languages.
IndicVoices	AI4Bharat	Spontaneous rather than read speech — the realistic condition.
Vaani	Google & IISc	District-level accent coverage; supports the inclusivity claim.
FLEURS	Google	Figures comparable with published results.
Common Voice	Mozilla	Supplementary validation.
IndicTTS database	IIT Madras	Voice training data where a gap exists.
LIMMITS / SYSPIN	IISc	Multi-speaker Indic synthesis corpora.

COMPLIANCE STATEMENT

No proprietary, closed-source or commercial voice-activation SDK is used anywhere in the system. Every model executes on the device. No network request is made at runtime. Two components carry restrictive licences — espeak-ng under GPL-3.0 and Meta MMS under CC-BY-NC — and both are disclosed here rather than left to be discovered.

Roadmap and team

Eight weeks. The governing rule is that the end-to-end loop closes before any effort is spent on model quality.

13.1 Schedule

WEEK	OBJECTIVE	DELIVERABLE
1	Capture and recognise	Skeleton application; AudioRecord into Silero VAD into Vosk Hindi; recognised text on screen. Entry-tier target handset acquired. Licence register opened.
2	Transport	Link interface, RFCOMM implementation, frame codec with CRC, unit-tested decoder. Typed text traverses two phones. Independent of any model.
3	Close the loop	Piper synthesis on the receiving device. Speech in, speech out, end to end. Baseline latency measured and recorded. The project is now de-risked.
4	Real models	IndicConformer exported and quantised; five languages integrated; WER harness operational; Odia gap assessed and assigned.
5	Modes and alerts	Push-to-talk with hardware key, half and full duplex, alert path with alarm-stream routing, wake lock and lock-screen delivery.
6	Latency and reach	Chunked synthesis, stabilised partials, adaptive endpointing. BLE and Wi-Fi transports. Encryption, replay window and relay. Thermal soak begins.
7	Coverage and hardening	Remaining languages; template tables; normalisation suites; noise pre-processing; field testing at range and under noise; four-device relay soak.
8	Evidence and rehearsal	Scorecard generation, latency HUD, documentation, LoRa bench demonstration if hardware permits, demonstration rehearsed to a fixed script.

13.2 Allocation

ROLE	OWNS	PRINCIPAL RISKS CARRIED
Models — recognition	core-asr, export, quantisation, biasing	T-02, T-05, T-07
Models — synthesis	core-tts, normalisation, voice selection	T-05
Audio engine	core-audio, VAD tiers, endpointing, noise	T-01, T-12
Transport	core-link, core-proto, security, relay	T-08, T-09, T-10, S-01, S-02
Application	UI, PTT, alerts, provisioning, packs	S-05, P-03
Evaluation	bench, datasets, scorecard, documentation	T-03, P-02

THE SCHEDULING RULE THAT MATTERS

Week three is the hinge. A working loop with mediocre models can be improved incrementally under any amount of time pressure. Excellent models with no loop cannot be demonstrated at all. Every competing team that fails will fail by inverting this order.

The demonstration

Seven minutes, rehearsed to a fixed sequence. Each step answers a question the jury is already holding.

#ACTION	QUESTION IT ANSWERS
1Both handsets on the table, aeroplane mode visibly enabled, Bluetooth only.	<i>Is this really offline?</i> Settled in three seconds, before any claim is made.
2Hold transmit; speak a Tamil sentence. The second handset speaks it in Tamil.	<i>Does the core loop work?</i>
3Display the byte counter: transmitted 45 bytes, equivalent audio 96 000 bytes, ratio 2 133×.	<i>What have you actually achieved?</i> The compression figure, measured live rather than asserted.
4Point to the permanent latency strip: STT 210 ms, link 40 ms, TTS 180 ms, total 780 ms.	<i>How fast is it?</i> Instrumentation on screen, not on a slide.
5Lock the second handset and set it to silent. Send an alert. It wakes and announces at full volume.	<i>Does the alert requirement work as specified?</i>
6Release push-to-talk. Hold an ordinary two-way conversation with no button.	<i>Does telephone mode work?</i>
7Switch the receiving handset to Hindi. Send a template alert in Tamil; it is announced in Hindi.	<i>What else does the architecture give you?</i> Cross-language operation with no translation model.
8If hardware permits: disable Bluetooth between the handsets; route through the LoRa pair across the room.	<i>Does this work on a real radio link?</i> The strongest single moment available.
9Close on the scorecard: application size, memory, idle CPU, WER per language at four SNRs, RTF, latency distribution — all measured on an entry-tier handset.	<i>Can you prove any of it?</i>

REHEARSAL DISCIPLINE

Pairing is the most common cause of demonstration failure. Devices are paired before the session begins, the QR path is the fallback, and a third pre-configured handset is kept ready. The script is fixed in week eight and not improvised.

Why this wins

Many teams will demonstrate speech crossing a Bluetooth link. Differentiation comes from four things that are difficult to assemble in the final week and straightforward if planned from the first.

The four differentiators

#DIFFERENTIATOR	WHY IT SEPARATES US
1 The compression argument, derived and measured	Most submissions will show a working pipeline. Few will present the bandwidth table demonstrating that audio <i>cannot</i> cross a LoRa or satellite link while iTantra can. That table is the reason an ISRO panel wrote this problem statement, and answering it directly is the strongest available position.
2 A real radio hop	Two handsets in aeroplane mode with Bluetooth disabled between them, communicating through an ESP32 and LoRa pair. Approximately fifteen hundred rupees of hardware and two hundred lines of firmware convert a phone-to-phone demonstration into a deployable system.
3 Evidence rather than claims	Word error rate per language at four signal-to-noise ratios, critical-term error with and without biasing, sustained real-time factor after thermal soak, idle CPU from a Perfetto trace, latency distribution from the instrumented application — all taken on an entry-tier handset. The rubric allocates eighty per cent to measurable properties; supplying measurements is simply answering the question asked.
4 Template codes yielding free cross-language operation	One byte per common message, spoken in whichever of the ten languages the receiver has selected. It compresses further than any competing approach and delivers translation with no translation model — an architectural consequence rather than an added feature.

Positioning

iTantra is not a speech-recognition demonstration with a Bluetooth link attached. It is a communications system that happens to use speech models as its codec. That framing is accurate, it matches the problem statement's own language — *neural transceiver radio access for low bitrate links* — and it is the framing under which the design decisions in this document are coherent rather than arbitrary.

*Speech goes in one end. Speech comes out the other.
In between, it is small enough to cross a link
that could never carry a voice.*

What a successful outcome looks like

- Two handsets, aeroplane mode, ten languages, sub-second end-to-end delivery.
- An installer under thirty megabytes with two languages resident and eight available on demand.
- Idle CPU below two per cent, verified by trace rather than asserted.
- Word error rate reported per language and per noise condition, with critical-term accuracy separated out.
- Alerts that wake a locked, silenced handset and announce at full volume.
- The same bytes crossing a real radio link, demonstrated live.

Frame specification

A.1 Message types

CODE	TYPE	SEMANTICS
0x1	TEXT	Recognised speech, packed or plain. Fire and forget.
0x2	ALERT	Priority traffic. Pre-empts the queue, acknowledged and retried, announced on the alarm stream.
0x3	ACK	Acknowledges a sequence number. Payload is the acknowledged value.
0x4	PTT_CTL	Floor seize and release. Drives the channel-busy indicator.
0x5	HEARTBEAT	Liveness, every two seconds. Carries link metrics.
0x6	TEMPLATE	Single-byte message identifier from the shared table.
0x7	POSITION	Eight-byte packed latitude and longitude.
0x8	AUDIO_FB	Opus fallback when recogniser confidence is below threshold. Wi-Fi transport only.

A.2 Language indices

IDX	LANGUAGE	SCRIPT	UNICODE BLOCK BASE
0	English	Latin	U+0041
1	Hindi	Devanagari	U+0900
2	Bengali	Bengali	U+0980
3	Marathi	Devanagari	U+0900
4	Telugu	Telugu	U+0C00
5	Tamil	Tamil	U+0B80
6	Gujarati	Gujarati	U+0A80
7	Kannada	Kannada	U+0C80
8	Malayalam	Malayalam	U+0D00
9	Odia	Odia	U+0B00

Each Indic block spans 128 code points, of which roughly seventy occur in ordinary running text. Subtracting the block base yields a single-byte representation directly, which is the mechanism behind Level 2 compression.

A.3 Flag bits

BIT	NAME	MEANING
7	FINAL	Complete utterance; safe to synthesise
6	PARTIAL	Stabilised prefix; receiver may begin early synthesis
5	ENCRYPTED	Payload is AES-GCM protected
4	FRAGMENT	More fragments follow – BLE and LoRa only
3	PACKED	Payload uses single-byte script packing
2	reserved	–
1–0	CONFIDENCE	0 low, 1 medium, 2 high, 3 template-matched

A.4 Reserved node identifiers

VALUE	MEANING
0x00	Unassigned – a device that has not completed provisioning
0x01	Group creator, conventionally the base station
0x02–0xFD	Assigned nodes
0xFE	Reserved for gateway or relay hardware
0xFF	Broadcast – all units on the channel

Reference implementation

The core paths, in the form they will take in the codebase.

B.1 Recognition ---

```
// constructed once, at foreground-service start
val recognizer = OnlineRecognizer(
    assetManager = assets,
    config = OnlineRecognizerConfig(
        featConfig = FeatureConfig(sampleRate = 16000, featureDim = 80),
        modelConfig = OnlineModelConfig(
            transducer = OnlineTransducerModelConfig(
                encoder = "models/hi/encoder.int8.onnx",
                decoder = "models/hi/decoder.onnx",
                joiner = "models/hi/joiner.onnx"
            ),
            tokens = "models/hi/tokens.txt",
            numThreads = 2,
            provider = "cpu"
        ),
        enableEndpoint = true,
        endpointConfig = EndpointConfig(),
        decodingMethod = "modified_beam_search",
        hotwordsFile = "models/hi/alert-lexicon.txt",
        hotwordsScore = 1.5f
    )
)

// per 100 ms of captured audio
stream.acceptWaveform(samples, 16000)
while (recognizer.isReady(stream)) recognizer.decode(stream)

val text = recognizer.getResult(stream).text
if (recognizer.isEndpoint(stream)) {
    link.send(Frame.encode(text, lang = HI, dst = BROADCAST))
    recognizer.reset(stream)
}
```

B.2 Synthesis with streaming playout

```
val tts = OfflineTts(assets, OfflineTtsConfig(
    model = OfflineTtsModelConfig(
        vits = OfflineTtsVitsModelConfig(
            model    = "models/hi/hi_IN-voice.onnx",
            tokens   = "models/hi/tokens.txt",
            dataDir  = "espeak-ng-data"
        ),
        numThreads = 2
    )
))

// clause-level chunking: playback starts on the first chunk
normalise(received, lang).splitClauses().forEach { chunk ->
    val audio = tts.generate(chunk, sid = speakerFor(frame.src), speed = 1.0f)
    track.write(audio.samples, 0, audio.samples.size, WRITE_BLOCKING)
}
```

B.3 Frame encoding

```
fun encode(text: String, lang: Int, dst: Int): ByteArray {
    val template = TemplateTable.match(text)          // Level 3
    val body = when {
        template != null -> byteArrayOf(template.id)
        else               -> ScriptPacker.pack(text, lang)    // Level 2
    }
    val payload = if (encrypted) aesGcm.seal(body, seq) else body

    return buildFrame {
        u8(MAGIC_V1)
        u8((if (template != null) TYPE_TEMPLATE else TYPE_TEXT) shl 4 or lang)
        u16(seq++)
        u8(flags)
        u16(payload.size)
        u8(myNode); u8(dst); u8(myGroup); u8(DEFAULT_TTL)
        bytes(payload)
        u16(crc16(soFar()))
    }
}
```

B.4 Receive path

```
link.incoming.collect { raw ->
    val f = Frame.decode(raw) ?: return@collect      // CRC failure
    if (!accept(f)) return@collect                  // not addressed to us
    if (!replayWindow.admit(f.src, f.seq)) return@collect // replay
    relayIfNeeded(f)

    val text = when (f.type) {
        TYPE_TEMPLATE -> TemplateTable.render(f.payload[0], myLanguage)
        else           -> ScriptPacker.unpack(f.payload, f.lang)
    }

    if (f.type == TYPE_ALERT) alertPlayer.announce(text, f.lang)
    else                      ttsPlayer.speak(text, f.lang)
}
```

B.5 Stream framing

```
fun InputStream.readFully(buf: ByteArray) {  
    var off = 0  
    while (off < buf.size) {  
        val n = read(buf, off, buf.size - off)  
        if (n < 0) throw EOFException()  
        off += n  
    }  
}
```

Glossary

TERM	DEFINITION
ASR / STT	Automatic speech recognition — converting audio into text.
TTS	Text to speech — converting text into an audio waveform.
VAD	Voice activity detection — deciding whether a short window of audio contains human speech.
Endpointing	Deciding that an utterance has finished, so that a final result can be emitted.
WER	Word error rate — insertions plus deletions plus substitutions, divided by reference word count. Lower is better.
RTF	Real-time factor — processing time divided by audio duration. Below 1.0 is faster than real time.
Mel spectrogram	A time-by-frequency representation of audio on a perceptually spaced frequency scale; the standard input to acoustic models.
Conformer	An acoustic model architecture combining convolution for local acoustic detail with self-attention for long-range context.
Transducer / CTC	Decoding schemes that map a sequence of frame predictions onto a shorter sequence of output tokens.
VITS	A synthesis architecture generating waveforms directly from text in a single pass, including duration modelling.
Quantisation	Re-encoding model weights at lower precision — typically 8-bit integer — for smaller size and faster inference.
ONNX	An open interchange format for neural networks, executable by ONNX Runtime on ARM.
Contextual biasing	Boosting decoding scores for a supplied phrase list, improving recognition of known domain vocabulary without retraining.
RFCOMM / SPP	The Bluetooth serial port profile — a virtual serial cable, and the same interface a radio module presents.
GATT	The Bluetooth Low Energy attribute protocol; data is exchanged through named characteristics rather than a stream.
MTU	Maximum transmission unit — the largest payload a single packet can carry.
LoRa	A long-range, low-bitrate radio modulation operating licence-free in India at 865–867 MHz.
TTL	Time to live — a hop counter that bounds how far a relayed frame may travel.
Replay window	A record of recently accepted sequence numbers, preventing a captured frame from being retransmitted by an attacker.
Half / full duplex	Half duplex permits one direction at a time — the walkie-talkie mode. Full duplex permits both simultaneously — the telephone mode.
Barge-in	Detecting that the local user has begun speaking while output is playing, and ducking or halting that output.
SNR	Signal-to-noise ratio in decibels; the standard way of specifying how adverse an acoustic test condition is.

